

Hakan Bora Yavuzkara

704-956-8760, hayavuzkara@davidson.edu, Davidson, NC, www.linkedin.com/in/hakan-bora-yavuzkara

EDUCATION

Davidson College

Davidson, NC

B.S. Physics & B.S. Computer Science

August 2024 - Expected May 2028

• GPA: 3.98

• James B. Duke Scholar (Full Tuition), Chidsey Leadership Fellow, Wolfram Student Ambassador

• Coursework: Computational Physics, Physics I&II (+lab), Data Structures, Programming in the Humanities (Mathematica), Linear Algebra, Classical Mechanics, Modern Physics, Discrete Structures, Computer Organization, Electricity & Magnetism, Calculus III, Quantum Chemistry, Software Design, Independent Physics Research (Deep Learning for Nuclear Physics)

American Robert College of Istanbul

Istanbul, Turkey

Turkish High School Diploma & AP Courses

August 2019 - June 2024

• GPA: 96.14

• Coursework: AP Calculus BC, AP Statistics, AP Physics C: Mechanics and Electricity & Magnetism, AP Psychology, AP Art History, Introduction to Modern Physics, Introduction to Electronics

AWARDS & HONORS

Davidson College Sophomore CS Award

May 2026

DataFest Davidson Best Use of External Data

April 2025

Hack@Davidson Best AI Chatbot via OpenAI API

Feb 2025

International Astronomy and Astrophysics (IAAC) Competition Silver Honor

Jul 2024

Kadir Has University “Advanced Electricity and Magnetism”, “Augmented Chemistry” and “Introduction to Quantum Mechanics” Outstanding Success

Aug 2024

ISBO Physics Olympiads 16th Place in Turkey

May 2023

Yusuf Durmuş Physics Camp High Honor Certificate (2nd Place)

Jul 2021

EXPERIENCE

Davidson College Data Cats

Davidson, NC

Consultant

Aug 2025-Present

• Provide data analytics and technical support in R, Python, and Mathematica to students

• Conduct one-on-one consultations and drop-in sessions, averaging 6 hours per week

Davidson College Center for Teaching and Learning

Davidson, NC

Physics Embedded Tutor/Computer Science Drop-in Tutor/Astronomy Grader

Feb 2025-Present

• Drop-in Tutor for Physics I&II and Computational Physics for 6 hours per week

• Selected as one of 4 Superhero tutors for the Spring 2026 semester for my work in Computational Physics.

• Embedded Tutor for Programming in the Humanities (Mathematica) class for 4 hours per week

• Grader for weekly homeworks in the Astronomy + Lab course for 4 hours per week

Davidson College ALPhA Research Group

Davidson, NC

Student Researcher

Jun 2025-Present

• Optimized the machine learning transformer models AdaPoinTr and Partial2Complete under the guidance of Dr. Michelle Kuchera and Dr. Ramanujan to reconstruct the Active Target Time Projection Chamber's recoded low-energy nuclear reaction tracks of $^{22}\text{Mg}+\alpha$ and $^{16}\text{O}+\alpha$ to improve the fitting algorithms for nuclear reactions

• Achieved F1 scores of .873 for $^{22}\text{Mg}+\alpha$ and .947 for $^{16}\text{O}+\alpha$, comparing the error between the model prediction and ground truth point clouds

• Research is ongoing, and the plan is for the final version to be used in the Facility for Rare Isotope Beams at Michigan State University.

S3++ Summer Science Program

Požega, Croatia

Student Researcher

Aug 2023-Sep 2023

• Participated in the “Quantum Two-Dimensional Matter: An Application of Graphene” group to build a graphene transistor and work on the theory part: solved the time-independent Schrödinger equation for a double finite barrier to replicate a graphene transistor, and programmed a code for the probability density of the electron around the system to see if it mimics a semiconductor.

• Presented our goal and findings through a presentation to the rest of the program and gave background information about our research.

Sabancı University Applied Nanotechnology Summer School

Istanbul, Turkey

Student Participant

Jul 2022-Aug 2022

• Studied nanotechnology and recent scientific discoveries at Sabancı University, taught by a different professor in every class.

• Made a battery, a super resistor, and conducted a redox reaction experiment.

• Visited one of the only cleanrooms in Turkey and learned the process of making nanomaterial through multiple demonstrations.

PUBLICATIONS & PRESENTATIONS

Poster Presentation "Transformers for Track Completion" SPSCon 2025	Nov 2025
Oral Presentation "Transformers for Track Completion" Division of Nuclear Physics Conference 2025	Oct 2025
Poster Presentation "Transformers for Track Completion" at Davidson College Fall Research Symposium	Sep 2025
Poster Presentation "Transformers for Track Completion" at FRIB in Michigan State University	Jul 2025

VOLUNTEER EXPERIENCE

Newtons of the Future	Istanbul, Turkey
<i>Founder, Co-Leader</i>	Sep 2022 - Dec 2022
• Co-led original project which explored the basics of physics with 5th and 6th graders in Piri Reis Secondary School • Explored physics with a 6 classes of 15, for 6 hours per day on weekends • Gathered 11 volunteers from my high school for each class to have 2, and explored with a class myself. • Created a curriculum that included a different physics subject (energy, motion, electricity, matter, pressure, etc.) each week • Ended each day with a general physics experiment relevant to the topic of the day, including springs, projectile motion, friction, or electric circuits	

CAMPUS INVOLVEMENT AND LEADERSHIP

Davidson College Society of Physics Students	Davidson, NC
<i>President</i>	Aug 2025-Present

- Lead SPS chapter of 210 members: promote physics events, SPS scholarships, and Graduate School Visits
- Organize Space Day (November 2025), and Star Party (February 2026)
- Volunteer at 2024-2025 Star Party (70+ attendees) and Ice-Cream Socials (40+ attendees)

Davidson College Hack@Davidson	Davidson, NC
<i>Logistics Team Member</i>	Aug 2025-Present

- Created PowerPoints that showed the strengths and weaknesses of individual players using advanced statistics

Davidson College Games Club	Davidson, NC
<i>Chess Team Co-Leader</i>	Aug 2025-Present

- Co-leading Davidson A chess team in PlayVS College League

Davidson College CATS' STATS	Davidson, NC
<i>Basketball Player Scout, Data Engineering Team Leader</i>	Oct 2024-Present

- Created PowerPoints that showed the strengths and weaknesses of individual players using advanced statistics

Robert College Astronomy Club	Davidson, NC
<i>Co-President (2023-2024)</i>	Sep 2020-May 2024

- Designed and printed the school's astronomy article, *Orion*, in 2022 and 2023.
- Helped build a solar filter for the school's telescope, later used for astrophotography.
- Proposed a project for a permanent telescope shed in the school campus.
- Led weekly gatherings based on discussing astronomy, doing introductory learning, and writing articles for *Orion*.

SKILLS

- Languages: Turkish (Native), English (Advanced Academic: DET 150/160, SAT 1510/1600), Japanese (Basic)
- Technical Skills: Java, Python, PyTorch, TensorFlow, Pandas, Mathematica, Git, Gaussian, R